

Mohamed Taha Abo Heiba

Cairo, Egypt | +20 114 971 1922 | mohamed-aboheiba@outlook.com | Portfolio: motahaaboheiba.github.io

LinkedIn: linkedin.com/in/mohamed-taha-abo-heiba | GitHub: github.com/MoTahaAboHeiba

PROFESSIONAL SUMMARY

Data Engineer with production experience building **batch ETL, streaming, and cloud-native lakehouse pipelines**, processing 200K+ records through **Medallion Architecture** systems on **Databricks, Spark, and AWS**. Also independently built and deployed a production **RAG API** achieving 79.45% HitRate@5 and 81.4% faithfulness. **AWS Certified Cloud Practitioner (CLF-C02)** and **Huawei HCIA-Big Data certified**; awarded **best-performing trainee** in Huawei Big Data Associate Program (NTI).

EXPERIENCE & TRAINING

- **AWS Cloud Engineer Trainee | National Telecommunication Institute (NTI)** *Oct 2025 – Dec 2025*
 - Architected multi-tier AWS app (**EC2/RDS/VPC/S3**) applying **Well-Architected Framework** to deliver fault-tolerant, production-ready infrastructure
 - Ensured **high availability** via **RDS Multi-AZ** configuration; reduced cold storage costs by implementing S3 lifecycle policies to Glacier
- **Big Data Engineer Trainee | National Telecommunication Institute (NTI)** *Jul 2025 – Sep 2025*
 - Awarded **best-performing trainee**; earned exam voucher and passed Huawei HCIA-Big Data certification on first attempt
 - Built GB-scale **Spark batch pipelines** on Hadoop (HDFS/YARN/Hive) and **Kafka+Flink streaming pipelines** across 15+ labs, operating distributed data systems at scale across **batch and real-time processing patterns**
- **Data Engineer Trainee | Digital Egypt Pioneers Initiative (DEPI)** *Apr 2024 – Oct 2024*
 - Automated **ETL workflows** by building 4 **SSIS packages** ingesting 7 CSV sources into **SQL Server via 17 stored procedures**, reducing manual effort and accelerating data delivery
 - Delivered 70K-record validated dataset powering 97.4% churn model to identify and target at-risk customer segments

PROJECTS

- **E-Commerce Lakehouse Databricks [GitHub](#)** | Databricks, PySpark, Delta Lake, Unity Catalog, Jobs
 - Built an end-to-end **cloud data lakehouse pipeline** processing 200K+ records from 6 data sources through a 3-task Databricks Jobs DAG (**Bronze** ingestion → **Silver** transformation → **Gold** star schema)
 - Delivered a governed, analytics-ready Delta Lake platform using **Medallion Architecture** and **Unity Catalog**, enabling downstream **OLAP consumption** via dimensional **star schema** modeling
- **SQL Data Warehouse Project [GitHub](#)** | SQL Server, T-SQL, Star Schema, Medallion Architecture
 - Architected a fully automated **warehouse pipeline** ingesting 60K+ records **across 6 sources via 2 master stored procedures**, applying **OLTP→OLAP** separation, **star schema** design, and idempotent **ETL** execution
 - Delivered **star schema** as SQL views with **per-step execution-time logging** for direct bottleneck identification
- **EduMate RAG [Live](#) | [GitHub](#)** | FastAPI, LangChain, Qdrant Cloud, ChromaDB, Huggingface Space, Groq
 - Independently built EduMate-RAG, a decoupled microservice (**platform-integrated + standalone modes**) with **hybrid BM25 + semantic search** over Qdrant Cloud in production and used ChromaDB for dev, indexing 24K+ document chunks via automated PDF pipeline; deployed on Hugging Face Spaces at ~240ms avg latency
 - Evaluated on 73 QA pairs: 79.45% HitRate@5, **81.4% faithfulness**, improved via reranking and refined prompting
- **Online Retail Analytics (Power BI & ETL Project) [GitHub](#)** | Power BI, Power Query, DAX
 - Cleaned 541K→406K records and engineered 8 features to build analytics-ready retail dataset for Power BI reporting
 - Built 3-page Report (30+ visuals, 5 DAX measures) surfacing top 25% of customers driving 65% of 9.73M revenue

CERTIFICATIONS

Huawei HCIA Big Data Associate | Huawei Technologies Co., Ltd

Issued: Mar 2026 | Valid: Mar 2029

AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services

Issued: May 2026 | Valid: May 2029

TECHNICAL SKILLS

- **Data Engineering & ETL Pipelines:** Medallion Architecture, Data Ingestion, (Batch, Stream Processing), Apache Airflow, ETL/ELT
- **Big Data & Distributed Computing:** PySpark, Spark SQL, Databricks, Hadoop (HDFS, YARN, Hive), Kafka, Flink, Delta Lake
- **Data Warehousing:** SQL Server, Dimensional Modeling (Star / Snowflake Schema), T-SQL, Stored Procedures, Unity Catalog
- **Databases:** PostgreSQL, MySQL, DynamoDB, HBase
- **Programming & DevOps:** Python (Pandas, NumPy), Bash, Docker, Git / GitHub
- **Cloud & AWS Data Services:** AWS (S3, EC2, RDS, VPC, IAM, Lambda, Glue, Redshift, Athena, EMR, ELB, Auto Scaling)
- **Analytics & Visualization:** Power BI, DAX, Power Query, Matplotlib, Pipeline Monitoring & Logging

EDUCATION

B.Sc. Computer Science — Faculty of Computers & Artificial Intelligence, Fayoum University

Oct 2022 – Jun 2026

GPA: 3.4 / 4.0 | **Core Coursework:** Big Data Engineering, Cloud Computing, Database Management Systems, Advanced SQL, Data Structures & Algorithms, Machine Learning, Operating Systems (Linux, Docker), Python Programming, Data Visualization (Power BI)